

WATKINSVILLE 5 SSE, GA, USA

WMO: 740060

Lat: 33.7837N

Lon: 83.3896W

Elev: 226

StdP: 98.64

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 63850

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.1	-3.0	-13.6	1.2	-2.1	-11.6	1.4	0.0	6.3	8.2	5.7	7.9	1.9	N/A	0.336

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	34.6	23.4	33.4	23.4	32.2	23.3	25.5	30.7	25.1	30.2	24.6	29.7	2.4	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.1	19.5	27.3	23.6	18.9	27.0	23.1	18.4	26.6	79.3	30.7	77.3	30.2	75.5	29.9	27.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.4	4.6	3.9	DB	-8.7	36.5	2.3	1.9	-10.4	37.9	-11.7	39.0	-13.0	40.0	-14.7	41.4	(4)
			WB	-9.8	26.4	2.5	0.5	-11.7	26.7	-13.1	27.1	-14.6	27.4	-16.4	27.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.9	6.6	8.6	12.3	16.6	21.0	25.1	26.2	25.8	23.1	17.3	11.5	8.5
	DBStd	7.97	5.38	4.96	4.75	3.84	3.35	2.20	1.83	2.05	2.73	4.15	4.25	4.75
	HDD10.0	372	134	79	32	3	0	0	0	0	0	2	33	88
	HDD18.3	1508	363	273	193	77	17	0	0	0	2	68	209	307
	CDD10.0	2897	29	40	103	201	343	452	501	490	392	229	76	41
	CDD18.3	992	0	1	6	24	101	202	242	232	144	37	2	1
	CDH23.3	8701	0	3	54	243	896	1866	2231	2004	1117	273	13	1
	CDH26.7	3504	0	0	4	37	292	819	1001	876	416	59	0	0
Wind	WSAvg	1.8	2.1	2.1	2.2	1.9	1.7	1.6	1.5	1.4	1.5	1.7	1.8	1.9
Precipitation	PrecAvg	1250	112	114	125	88	83	113	113	112	105	78	94	112
	PrecMax	1653	177	215	236	236	227	297	256	241	325	233	239	310
	PrecMin	821	62	42	31	33	26	34	36	25	1	0	17	49
	PrecStd	228	32	41	52	47	40	62	64	59	71	57	53	59
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.6	23.5	27.3	29.6	33.6	35.8	35.6	36.5	33.8	31.1	25.2	22.6
		MCWB	15.8	17.6	17.1	19.4	21.2	23.1	23.9	23.0	22.6	21.8	17.9	17.7
	2%	DB	19.2	20.7	24.9	27.7	31.4	34.0	34.2	34.4	32.3	28.0	22.8	20.5
		MCWB	14.4	15.4	16.6	18.9	20.8	23.4	24.1	23.2	22.5	19.9	15.7	17.4
	5%	DB	17.3	18.8	22.7	26.2	29.9	32.6	33.1	32.7	30.9	26.4	20.8	18.5
		MCWB	13.9	14.6	15.5	17.9	20.8	23.3	24.1	23.6	22.3	19.0	15.5	15.9
	10%	DB	15.3	16.9	20.6	24.6	28.2	31.2	31.8	31.2	29.3	24.5	19.0	16.5
		MCWB	12.3	13.3	14.1	17.1	20.5	23.1	23.9	23.5	22.0	18.3	14.6	13.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.7	18.9	19.0	21.6	23.5	25.6	26.0	26.0	24.9	23.9	20.3	20.1
		MCDB	19.9	21.7	24.1	26.1	28.5	31.3	31.6	31.1	30.0	27.8	22.0	21.4
	2%	WB	16.4	17.7	18.0	20.4	22.7	25.0	25.5	25.5	24.1	22.6	18.6	18.4
		MCDB	18.4	20.3	22.8	25.2	27.8	30.9	31.1	30.1	29.6	25.8	20.8	19.6
	5%	WB	14.8	16.2	17.0	19.6	22.0	24.4	25.1	25.0	23.6	21.4	17.1	16.4
		MCDB	16.5	18.4	21.2	24.0	27.0	30.4	30.7	29.4	28.4	24.5	18.9	17.7
	10%	WB	12.6	14.5	15.8	18.6	21.4	23.8	24.6	24.4	22.9	19.9	15.3	14.2
		MCDB	14.5	16.5	19.3	23.0	26.1	29.6	30.0	28.9	27.2	23.2	18.0	16.1
Mean Daily Temperature Range	5% DB	MDBR	10.5	10.6	11.6	12.4	12.0	11.2	10.7	10.3	10.7	11.5	11.6	9.9
		MCDBR	11.7	11.9	13.4	14.1	14.1	13.2	12.4	12.6	12.9	13.3	13.5	11.1
		MCWBR	8.1	7.1	6.6	6.3	5.3	4.3	3.7	4.2	4.6	6.0	7.8	7.3
	5% WB	MCDBR	9.9	9.4	11.2	11.3	11.0	11.2	10.8	10.0	10.6	10.4	9.6	8.6
		MCWBR	8.4	6.9	6.4	6.1	4.8	4.4	4.0	3.8	4.2	5.6	7.1	7.2
Clear-Sky Solar Irradiance	taub	0.314	0.319	0.355	0.389	0.450	0.480	0.520	0.518	0.423	0.383	0.342	0.311	
	taud	2.521	2.499	2.425	2.317	2.185	2.150	2.058	2.049	2.299	2.390	2.456	2.555	
	Ebn at Noon	902	934	920	899	843	814	779	773	839	848	857	886	
	Edh at Noon	83	94	110	128	148	153	167	166	123	104	88	76	
All-Sky Solar Radiation	RadAvg	2.74	3.33	4.41	5.50	6.09	6.27	5.97	5.47	4.75	4.02	3.04	2.34	
	RadStd	0.15	0.40	0.32	0.37	0.48	0.48	0.41	0.28	0.46	0.47	0.29	0.24	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (41 neighbors)		+0.54	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+170	+113

Nomenclature: See separate page